

ZHENGHANG XU

105 St. George Street, Rotman School of Management, University of Toronto
+1-437-435-0396 ♦ zhenghang.xu@rotman.utoronto.ca
personal website: <https://zhenghangxu-baymax.github.io/>

RESEARCH INTERESTS

AI for Queueing Discovery and Control

I develop AI and learning methods to understand queueing systems from data, evaluate how policy changes affect their performance, and improve decisions such as capacity allocation and pricing.

Methodologies	Causal Machine Learning, Queueing Theory, Optimal Control, Online Learning
Applications	Healthcare Management, Online Platforms, Inventory Management

EMPLOYMENT

Rotman School of Management, University of Toronto Postdoctoral Research Fellow Advisors: Nasser Barjesteh, Sheng Liu, Vahid Roshanaei	<i>2026 - Present</i>
---	-----------------------

EDUCATION

Rotman School of Management, University of Toronto Ph.D. in Operations Management and Statistics Advisors: Opher Baron, Dmitry Krass, Philipp Afèche	<i>2021 - 2026</i>
The Chinese University of Hong Kong, Shenzhen (CUHKSZ) B.Sc. in Statistical Science Rank: 1/151	<i>2017 - 2021</i>

PUBLICATIONS AND PAPERS UNDER REVISION

Queueing Causal Models: Comparative Analytics in Queueing Systems

with Opher Baron, Dmitry Krass, Mark van der Laan and Arik Senderovich

M&SOM, 2025 [Available at SSRN 5104446]

- Next-Gen Scholar's Symposium, 2026
- **First prize**, CORS 2025 Queueing and Applied Probability SIG
- **Winner**, 2024 Oded Berman Student Paper Competition
- **Finalist**, CORS 2024 Best Student Paper Competition (Open Category)
- Service Management SIG, M&SOM 2024

Managing Scarce MRI Capacity in Overloaded Queueing Systems

with Andre Cire, Adam Diamant, Eugene Furman and Opher Baron

Major revision at M&SOM, 2026 [Available at SSRN 6158086]

- **Finalist**, CORS 2026 Healthcare SIG

Leveraging Advanced Analytics to Streamline the Emergent Dialysis Process at Parkland Hospital

with Olga Bountali, Sila Cetinkaya, Michael Hahsler, Farnaz Nourbakhsh and Henry Quinones

Accepted at Healthcare Analytics [Available at SSRN 5066305]

A High-Fidelity, Machine-Learning Enhanced Queueing Network Simulation Model for Hospital Ultrasound Operations.

with Yihan Pan, Jin Guang, Chengwenjian Wang, Xuanming Zhang, Xinyun Chen, J.G. Dai, Yichuan Ding, Pengyi Shi, Hongxin Pan, Kai Yang and Song Wu

In 2021 *Winter Simulation Conference (WSC)* (pp. 1-12). IEEE.

WORKING PAPERS

Learning to Price in Queues: Arrival Observability and Value Randomness

with Philipp Afèche, Opher Baron and Dmitry Krass

To be submitted, draft available upon request

- 2024 Rotman TD MDAL Research Grant

Structural Causal Inventory Models: Identification and Policy Coverage

with Opher Baron and Dmitry Krass

Work in progress

Queueing Causal Models for Emergency Department Efficiency

with Opher Baron, Dmitry Krass, Mark van der Laan and Yufeng Zheng

Work in progress

- 2025 Rotman TD MDAL Research Grant

Fairness in Multi-Stage Healthcare Access: A Study of Congestion, Chronic Patients, and Operational Redesign

with Olga Bountali, Sila Cetinkaya, Michael Hahsler, Farnaz Nourbakhsh and Henry Quinones

Work in progress

HONORS AND AWARDS

First Prize , CORS Queueing and Applied Probability SIG	2025
Winner , Oded Berman Student Paper Competition	2024
Finalist , CORS Best Student Paper Competition (Open Category)	2024
Rotman TD MDAL Research Grant (\$15K)	2023, 2024, 2025
Ph.D. Fellowship, Rotman School of Management	2021 - 2026
Presidential Award for Outstanding Students (10/900+), CUHKSZ	2021
Undergraduate Research Assistantship, CUHKSZ	2019
First Class Academic Scholarship (top 3%), CUHKSZ	2019, 2020
Third Class Academic Scholarship (top 8%), CUHKSZ	2018
Dean's List, CUHKSZ	2018, 2019, 2020

PROFESSIONAL SERVICES

Session Chair, CORS	2025
Host a session under Queueing and Applied Probability SIG.	
Co-organizer, Rotman Young Scholar Seminar	2025 - present
Co-host weekly seminars and invite five speakers.	
Journal Referee for <i>Operations Research</i> , <i>Queueing Systems</i> , <i>INFOR</i> .	
Conference Referee for <i>M&SOM Main Conference</i> .	

INVITED TALKS & CONFERENCE PRESENTATIONS

Queueing Causal Models: Comparative Analytics in Queueing Systems

* denotes presentations given jointly with another speaker. This talk was presented by other speakers as a keynote and at several other invited seminars nationally and internationally.

Invited Seminars

- Next-Gen Scholar's Symposium, National University of Singapore *August 2026*
- School of Economics and Management, Tsinghua University *Dec 2025*
- College of Business, Shanghai University of Finance and Economics *Dec 2025*
- Antai College of Economics and Management, Shanghai Jiao Tong University *Dec 2025*
- College of Business, City University of Hong Kong *Dec 2025*
- Institute of Supply Chain Analytics, Dongbei University of Finance and Economics *Dec 2025*
- School of Management, Fudan University* *Sep 2024*
- Antai College of Economics and Management, Shanghai Jiao Tong University* *Sep 2024*
- School of Business, Renmin University of China* *Sep 2024*
- School of Management, Hefei University of Technology* *Sep 2024*
- School of Management and Economics, The Chinese University of Hong Kong, Shenzhen* *Sep 2024*
- Department of Decisions, Operations and Technology, CUHK Business School* *Sep 2024*
- IEDA, Hong Kong University of Science and Technology* *Sep 2024*
- Department of Logistics and Maritime Studies, The Hong Kong Polytechnic University* *Sep 2024*

Conference Presentations

- INFORMS Applied Probability Society *July 2025*
- Matrix-Analytic Methods (MAM) 12 Conference (Keynote)* *June 2025*
- CORS Queueing and Applied Probability SIG *June 2025*
- INFORMS Annual Meeting *Oct 2024*
- CORS Student Paper Competition (Open Category) *June 2024*
- Rotman TD MDAL Research Round Table *May 2024*
- INFORMS Annual Meeting *Oct 2023*
- CORS Annual Meeting *June 2023*

Learning to Price in Queues: Arrival Observability and Value Randomness

- Rotman TD MDAL Research Round Table *May 2026*
- POMS-HK Conference *Jan 2026*
- INFORMS Annual Meeting *Oct 2025*
- CanQueue workshop *August 2025*
- Prof. Huan Zheng's Group, Shanghai Jiao Tong University (Online) *July 2025*
- CORS Annual Meeting *June 2025*
- INFORMS Toronto Student Chapter Research Talk Series *May 2025*

Managing Scarce MRI Capacity in Overloaded Queueing Systems

- MSOM Conference *July 2026*
- CORS Healthcare SIG *June 2026*

Queueing Causal Models for Emergency Department Efficiency

- Rotman TD MDAL Research Round Table *May 2025*

TEACHING EXPERIENCES

Guest Speaker, Rotman School of Management, University of Toronto

RSM3090: Queueing Theory - Models and Analysis *Fall 2025*

Teaching Assistant, Rotman School of Management, University of Toronto

RSM1240: Statistics for Management (MBA) *Winter 2024, 2025*

RSM1282: Operations Management (MBA) *Winter 2025*

RSM8414: Tools for Probabilistic Models and Prescriptive Analytics (MMA) *Fall 2024*

RSM8432: Management Analytics Practicum (MMA) *Winter 2024*

RSM2408: Modeling and Optimization for Decision Making (MMA) *Fall 2023*

RSM8415: Service Analytics for Management Analysis (MMA)	<i>Winter 2023</i>
RSM2405: Supply Chain Management (MBA)	<i>Fall 2022</i>
Project Coach, Master of Management Analytics (MMA) Program, Rotman School of Management, University of Toronto	
Host companies: Imperial Oil, Ontario Health, Scotiabank	<i>Winter 2024</i>
Teaching Assistant, CUHKSZ	
BIO2001: General Biology (Undergraduate)	<i>Spring 2019</i>
CHM1001: General Chemistry (Undergraduate)	<i>Fall 2019</i>
MAT2040: Linear Algebra (Undergraduate)	<i>Spring 2020</i>
STA2002: Probability and Statistics II (Undergraduate)	<i>Spring 2021</i>

REFERENCES

Prof. Opher Baron

Operations Management & Statistics
Rotman School of Management
University of Toronto
Email: opher.baron@rotman.utoronto.ca

Prof. Dmitry Krass

Operations Management & Statistics
Rotman School of Management
University of Toronto
Email: dmitry.krass@rotman.utoronto.ca

Prof. Philipp Afèche

Operations Management & Statistics
Rotman School of Management
University of Toronto
Email: philipp.afeche@rotman.utoronto.ca

Last updated on October 5, 2026